



US 20250282122A1

(19) **United States**

(12) **Patent Application Publication**
YANG et al.

(10) **Pub. No.: US 2025/0282122 A1**

(43) **Pub. Date: Sep. 11, 2025**

(54) **LAMINATED GLASS AND VEHICLE**

C03C 17/34 (2006.01)

C03C 17/36 (2006.01)

(71) Applicant: **FUYAO GLASS INDUSTRY GROUP CO., LTD.**, Fuqing (CN)

(52) **U.S. Cl.**

CPC .. B32B 17/10229 (2013.01); **B32B 17/10036**

(2013.01); **B32B 17/1011** (2013.01); **B32B**

17/10201 (2013.01); **B32B 17/10449**

(2013.01); **B32B 17/10495** (2013.01); **B32B**

17/10651 (2013.01); **B32B 17/10761**

(2013.01); **B60J 1/001** (2013.01); **B60J 3/007**

(2013.01); **C03C 17/3435** (2013.01); **C03C**

17/3626 (2013.01); **C03C 17/3644** (2013.01);

C03C 17/3657 (2013.01); **B32B 2307/304**

(2013.01); **B32B 2605/00** (2013.01); **C03C**

2217/948 (2013.01); **C03C 2218/156** (2013.01)

(21) Appl. No.: **19/216,117**

(22) Filed: **May 22, 2025**

Related U.S. Application Data

(63) Continuation of application No. PCT/CN2023/133516, filed on Nov. 23, 2023.

Foreign Application Priority Data

Nov. 23, 2022 (CN) 202211472091.X

Publication Classification

(51) **Int. Cl.**

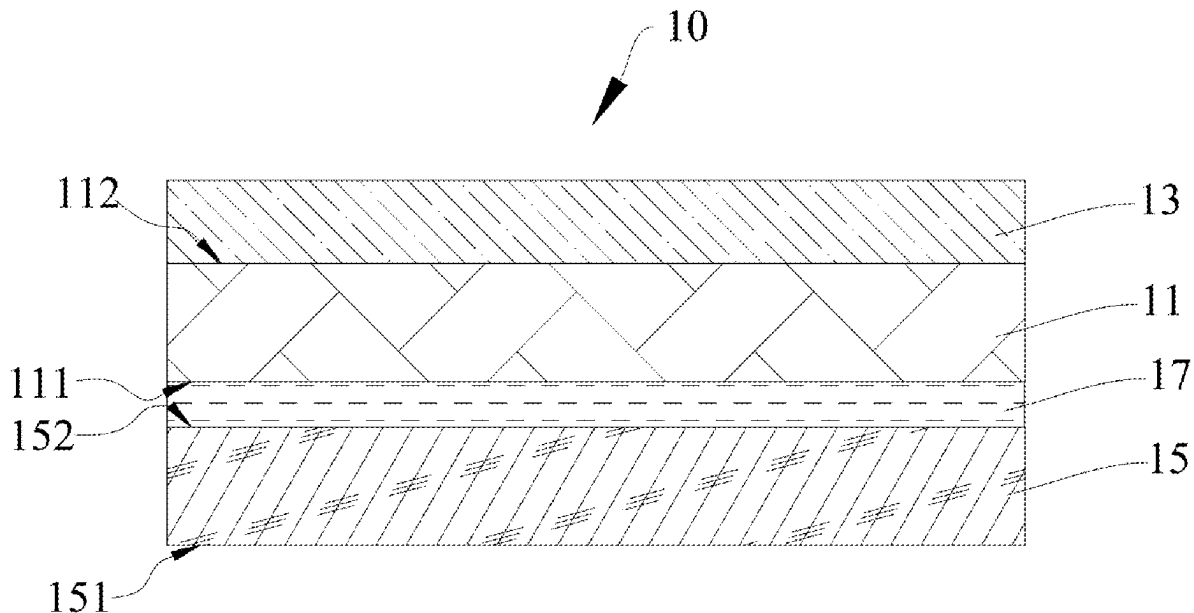
B32B 17/10 (2006.01)

B60J 1/00 (2006.01)

B60J 3/00 (2006.01)

ABSTRACT

A laminated glass and a vehicle are provided. The laminated glass includes an outer glass plate, an adhesive layer, an inner glass plate, and a low-emissivity coating stacked in sequence. The outer glass plate has a first surface and a second surface opposite to each other, the inner glass plate has a third surface and a fourth surface opposite to each other, and the adhesive layer is used to connect the second surface and the third surface. The low-emissivity coating is disposed on the fourth surface of the inner glass plate and includes 1 to 3 transparent conductive oxide coatings. A thermal insulation index C of the laminated glass is calculated according to the equation: $C = T_{nIR} / R_{IR}$, where the range of the thermal insulation index C is $0.008 \leq C \leq 0.4$.



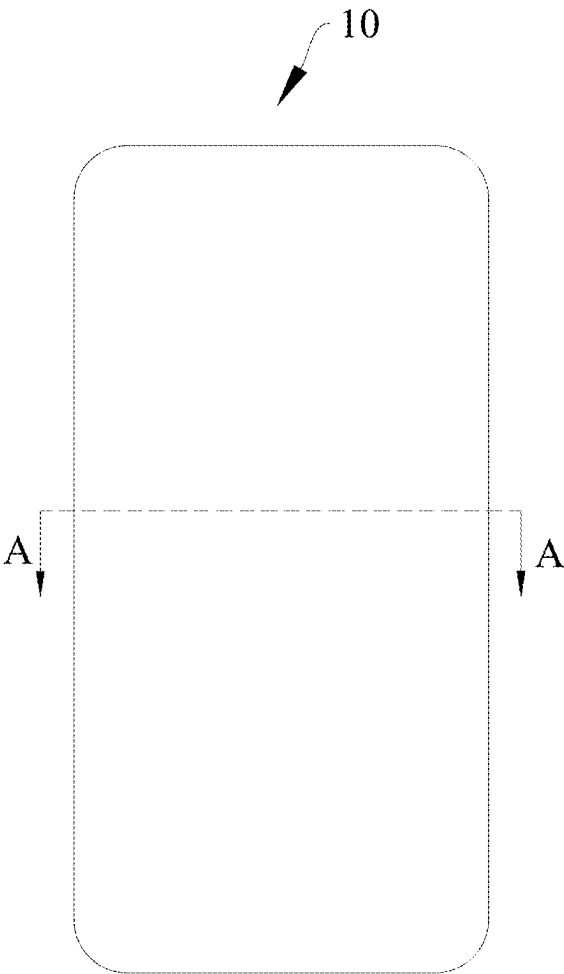


FIG. 1

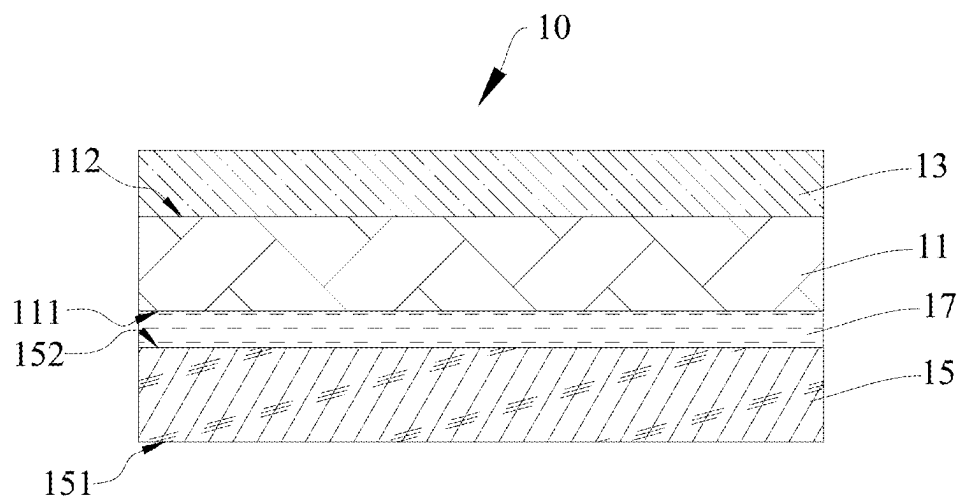


FIG. 2

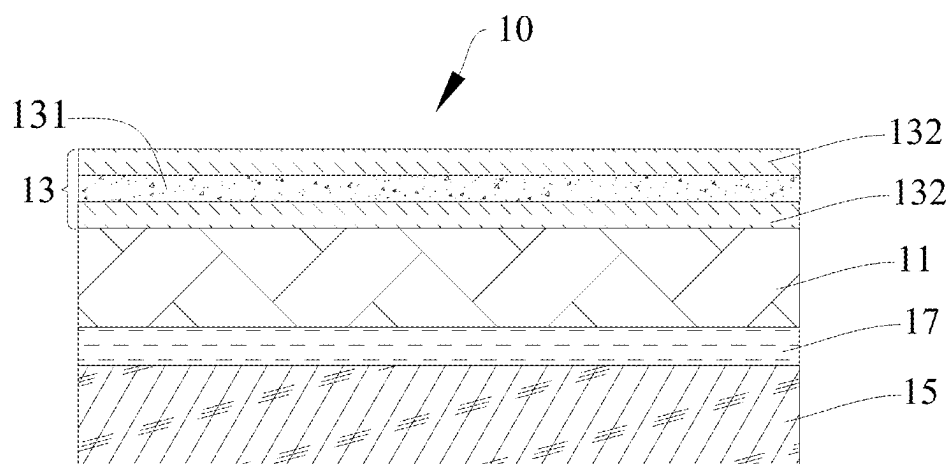


FIG. 3

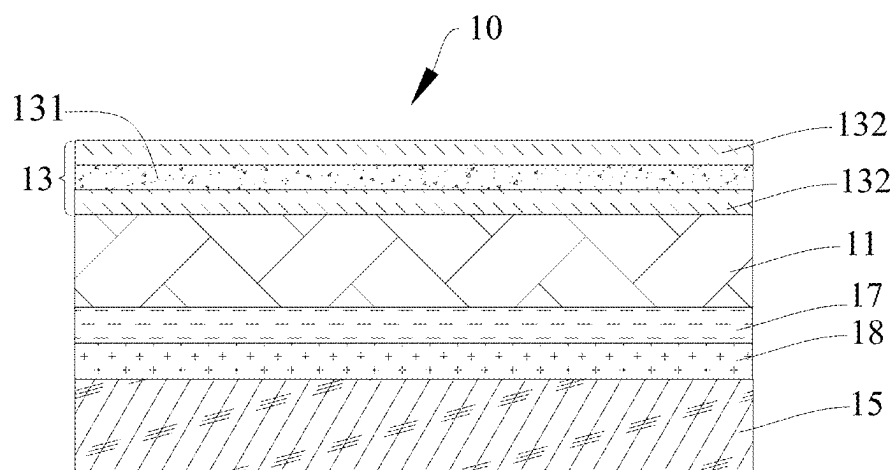


FIG. 4

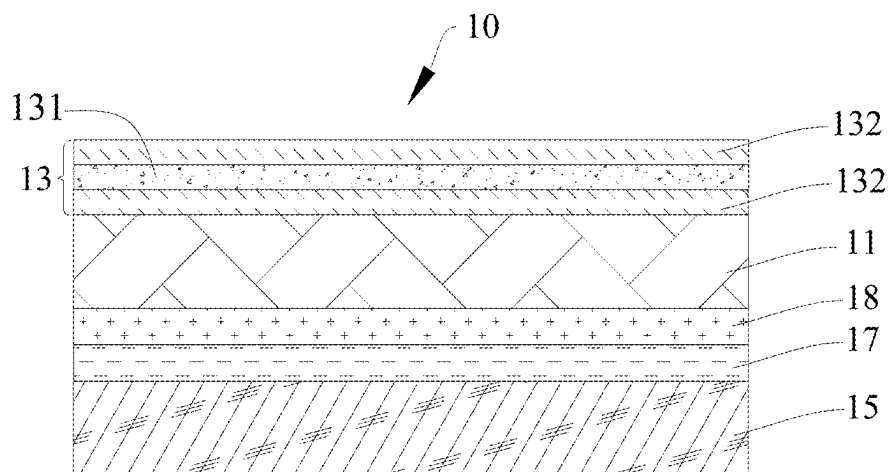


FIG. 5

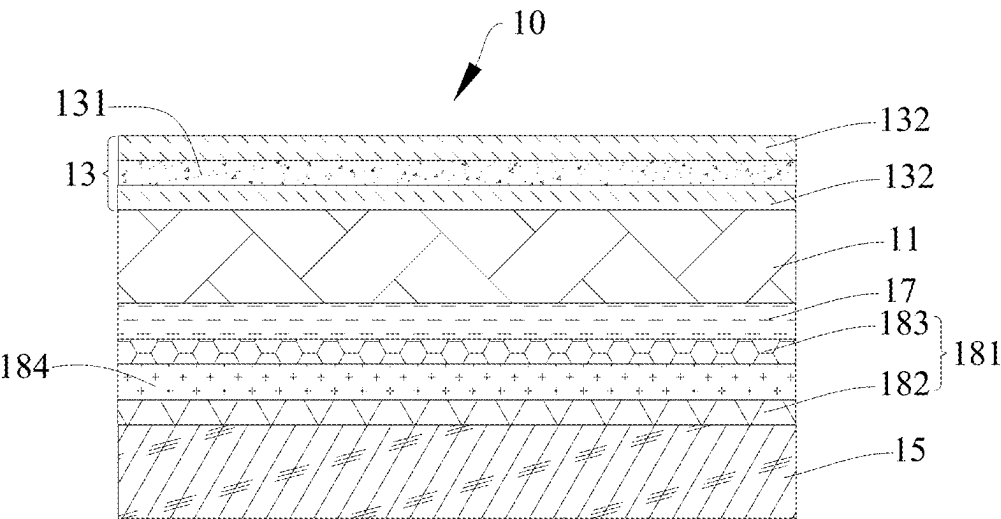


FIG. 6

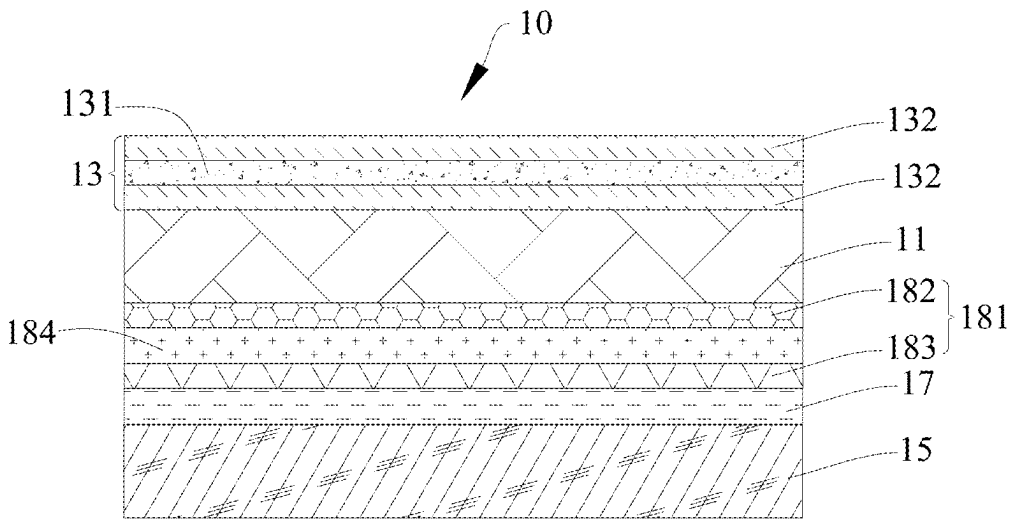


FIG. 7

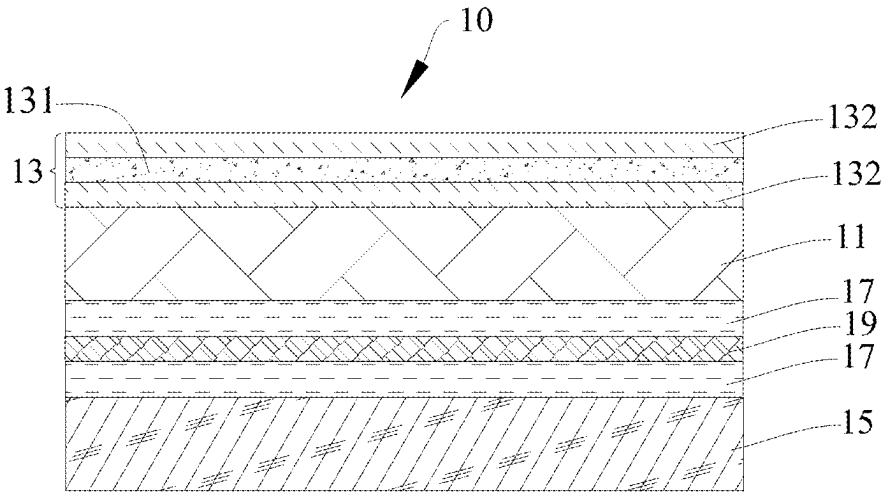


FIG. 8

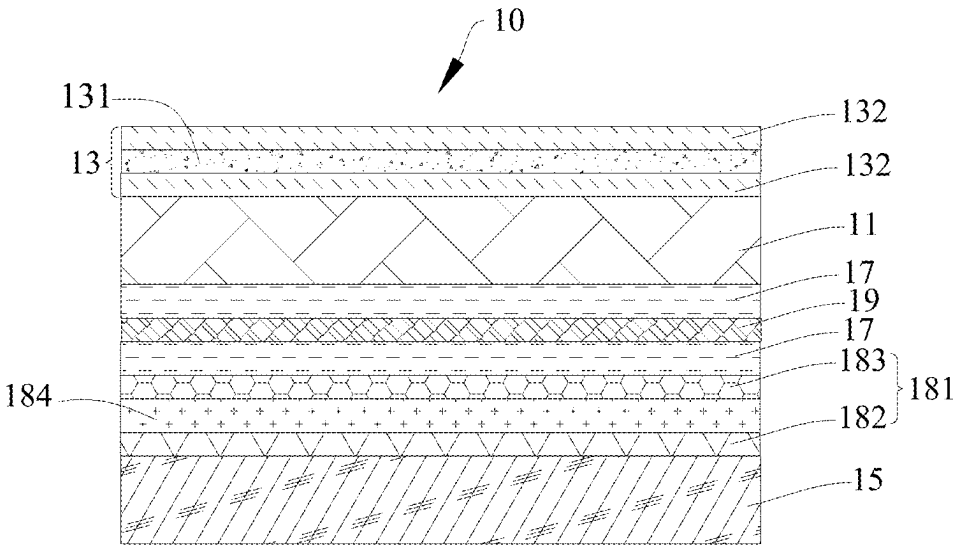


FIG. 9

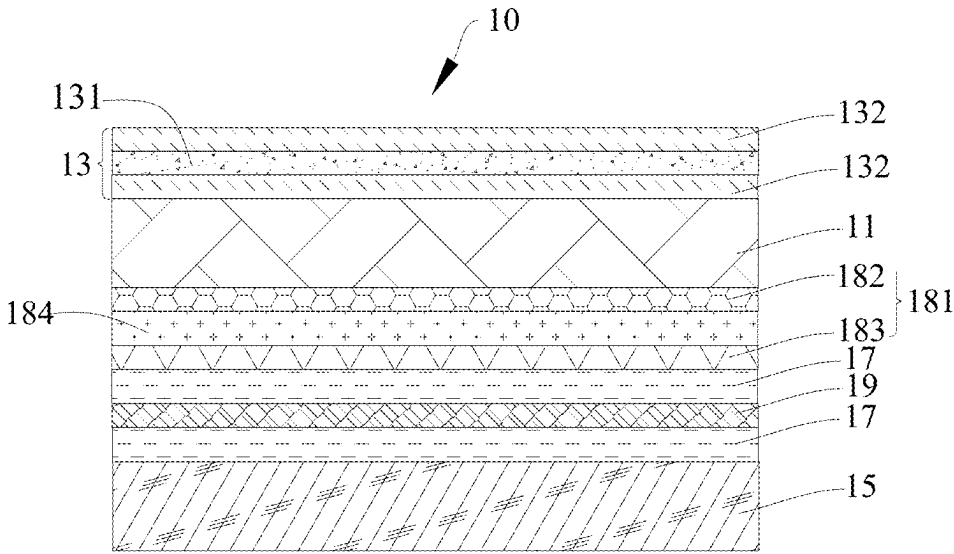


FIG. 10

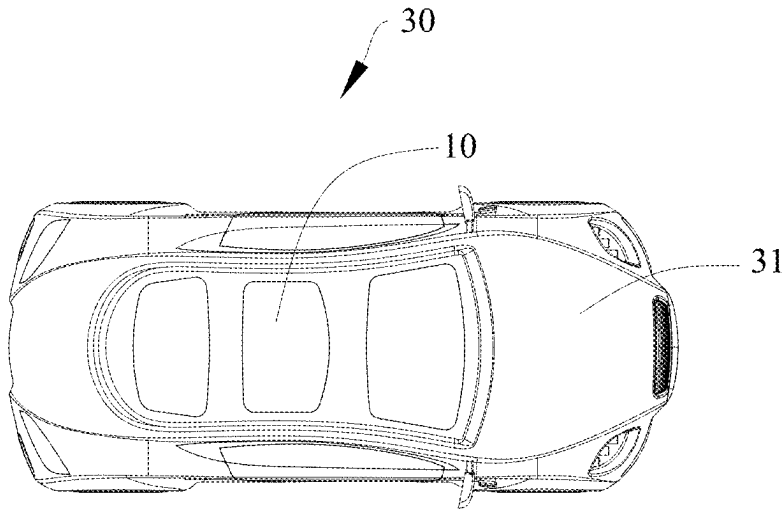


FIG. 11

LAMINATED GLASS AND VEHICLE

[0001] This application is a continuation of International Application No. PCT/CN2023/133516, filed Nov. 23, 2023, which claims priority to Chinese Patent Application No. 202211472091.X, filed Nov. 23, 2022, the entire disclosures of both of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to the field of glass, and in particular, to a laminated glass and a vehicle.

BACKGROUND

[0003] The interior space of a vehicle becomes extremely hot in summer due to high-temperature environment and strong direct sunlight, and becomes extremely cold in winter due to heat loss of the vehicle caused by low external temperature. More and more vehicles are using a sunroof glass coated with a coating having low-emissivity property on a surface close to the interior space, so as to achieve a “warm in winter and cool in summer” effect inside the vehicles. In summer, the low-emissivity coating can reduce the amount of long-wave thermal radiation emitted from the heated sunroof glass into the interior space of the vehicle, thereby preventing heat from entering the interior space of the vehicle. In winter, the low-emissivity coating can reduce the heat radiation from the interior space of the vehicle to the external environment, thereby preventing heat loss of the interior space of the vehicle.

[0004] As the sunroof glass is getting bigger, especially the sunroof glass or the panoramic roof glass on electric vehicles, traditional glass with low-emissivity coating as the sunroof glass is difficult to meet the requirement for thermal comfort inside the vehicle. Research has found that human skin is most sensitive to heat with a wavelength of 1000 nm to 1250 nm. In practical applications, the thermal blocking effect of the traditional sunroof glass is relatively poor, which fails to provide passages with an adequately comfortable sensation on their skin. Therefore, a sunshade needs to be used in combination with the sunroof glass, which fails to meet requirements of original equipment manufacturers (OEM), such as lightweight of vehicles, increasing headroom, reducing costs.

SUMMARY

[0005] The disclosure provides a laminated glass, including an outer glass plate, an adhesive layer, an inner glass plate, and a low-emissivity coating stacked in sequence. The outer glass plate has a first surface and a second surface opposite to each other, the inner glass plate has a third surface and a fourth surface opposite to each other, and the adhesive layer is used to connect the second surface and the third surface. The low-emissivity coating is disposed on the fourth surface of the inner glass plate and includes 1 to 3 transparent conductive oxide coatings. A thermal insulation index C of the laminated glass is calculated according to the equation: $C = T_{nIR} / R_{IR}$, where a range of the thermal insulation index C is $0.008 \leq C \leq 0.4$. T_{nIR} is a transmittance of the laminated glass for infrared with a wavelength of 1000 nm to 1250 nm, and R_{IR} is a reflectivity of the laminated glass for infrared with a wavelength of 780 nm to 2500 nm measured at one side of the low-emissivity coating away from the inner glass plate.

[0006] The disclosure further provides a vehicle. The vehicle includes a vehicle body and the laminated glass provided in the disclosure. The laminated glass includes an outer glass plate, an adhesive layer, an inner glass plate, and a low-emissivity coating stacked in sequence. The laminated glass is mounted on the vehicle body as at least one of a sunroof glass, a side window glass, or a rear windshield glass of the vehicle. The outer glass plate has a first surface and a second surface opposite to each other, the inner glass plate has a third surface and a fourth surface opposite to each other, and the adhesive layer is used to connect the second surface and the third surface. The low-emissivity coating is disposed on the fourth surface of the inner glass plate and includes 1 to 3 transparent conductive oxide coatings. A thermal insulation index C of the laminated glass is calculated according to the equation: $C = T_{nIR} / R_{IR}$, where a range of the thermal insulation index C is $0.008 \leq C \leq 0.4$, T_{nIR} is a transmittance of the laminated glass for infrared with a wavelength of 1000 nm to 1250 nm, and R_{IR} is a reflectivity of the laminated glass for infrared with a wavelength of 780 nm to 2500 nm measured at one side of the low-emissivity coating away from the inner glass plate.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] In order to describe technical solutions of embodiments of the disclosure more clearly, the following will give a brief introduction to the accompanying drawings used for describing the embodiments. Apparently, the accompanying drawings hereinafter described are some embodiments of the present disclosure. Based on these drawings, those of ordinary skill in the art can also obtain other drawings without creative effort.

[0008] FIG. 1 is a schematic structural view of a laminated glass provided in an embodiment of the disclosure.

[0009] FIG. 2 is a sectional view of a laminated glass provided in an embodiment of the disclosure, taken along line A-A in FIG. 1.

[0010] FIG. 3 is a sectional view of a laminated glass provided in another embodiment of the disclosure, taken along line A-A in FIG. 1.

[0011] FIG. 4 is a sectional view of a laminated glass provided in another embodiment of the disclosure, taken along line A-A in FIG. 1.

[0012] FIG. 5 is a sectional view of a laminated glass provided in another embodiment of the disclosure, taken along line A-A in FIG. 1.

[0013] FIG. 6 is a sectional view of a laminated glass provided in another embodiment of the disclosure, taken along line A-A in FIG. 1.

[0014] FIG. 7 is a sectional view of a laminated glass provided in another embodiment of the disclosure, taken along line A-A in FIG. 1.

[0015] FIG. 8 is a sectional view of a laminated glass provided in another embodiment of the disclosure, taken along line A-A in FIG. 1.

[0016] FIG. 9 is a sectional view of a laminated glass provided in another embodiment of the disclosure, taken along line A-A in FIG. 1.

[0017] FIG. 10 is a sectional view of a laminated glass provided in another embodiment of the disclosure, taken along line A-A in FIG. 1.

[0018] FIG. 11 is a schematic structural view of a vehicle provided in an embodiment of the disclosure.

Reference Signs

[0019] 10—laminated glass, 11—inner glass plate, 111—third surface, 112—fourth surface, 13—low-emissivity coating, 131—transparent conductive oxide coating, 132—first dielectric layer, 15—outer glass plate, 151—first surface, 152—second surface, 17—adhesive layer, 18—infrared reflection coating, 181—second dielectric layer, 182—second inner dielectric layer, 183—second outer dielectric layer, 184—metal layer, 19—light-adjustment component, 30—vehicle, 31—vehicle body.

DETAILED DESCRIPTION

[0020] The following will clearly and completely illustrate technical solutions of embodiments of the disclosure with reference to the accompanying drawings of embodiments of the disclosure. Apparently, embodiments described herein are merely some embodiments, rather than all embodiments, of the disclosure. Based on the embodiments of the disclosure, all other embodiments obtained by those of ordinary skill in the art without creative effort shall fall within the protection scope of the disclosure.

[0021] The terms “first”, “second”, and the like in the description, claims of the present disclosure, and the above accompanying drawings are used for distinguishing different objects, rather than for describing a specific order. In addition, the terms “include”, “have”, and any variations thereof are intended to cover non-exclusive inclusions. For example, a process, method, system, product, or apparatus that includes a series of steps or units is not limited to the listed steps or units, but optionally further includes steps or units not listed, or optionally further includes other steps or units inherent to the process, method, product, or apparatus.

[0022] Reference to “embodiment” or “implementation” herein means that a particular feature, structure, or characteristic described in conjunction with the embodiment may be included in at least one embodiment of the present disclosure. The presence of the term at each place in the specification does not necessarily refer to the same embodiment, nor does it refer to a separate or alternative embodiment that is mutually exclusive of other embodiments. It may be understood by those skilled in the art, both explicitly and implicitly, that the embodiments described herein may be combined with other embodiments.

[0023] Reference is made to FIG. 1 to FIG. 3. Embodiments of the disclosure provide a laminated glass 10. The laminated glass 10 includes an outer glass plate 15, an adhesive layer 17, an inner glass plate 11, and a low-emissivity coating 13 stacked in sequence. The outer glass plate 15 has a first surface 151 and a second surface 152 opposite to each other, the inner glass plate 11 has a third surface 111 and a fourth surface 112 opposite to each other, and the adhesive layer 17 is used to connect the second surface 152 and the third surface 111. The low-emissivity coating 13 is disposed on the fourth surface 112 of the inner glass plate 11 and includes 1 to 3 transparent conductive oxide coatings 131. A thermal insulation index C of the laminated glass 10 is calculated according to the equation: $C = T_{nIR} / R_{IR}$, where a range of the thermal insulation index C is $0.008 \leq C \leq 0.4$. T_{nIR} is a transmittance of the laminated glass 10 for infrared with a wavelength of 1000 nm to 1250 nm, and R_{IR} is a reflectivity of the laminated glass 10 for

infrared with a wavelength of 780 nm to 2500 nm measured at one side of the low-emissivity coating 13 away from the inner glass plate 11.

[0024] The laminated glass 10 provided in the disclosure includes the low-emissivity coating 13, which is disposed on the fourth surface 112 of the inner glass plate 11. When the low-emissivity coating 13 is mounted on a vehicle, the low-emissivity coating 13 can reduce the emissivity of the laminated glass 10 measured at the interior side of the vehicle. The low-emissivity coating 13 not only blocks the transmission of heat from the outside of the vehicle to the interior space of the vehicle, but also blocks the dissipation of heat from the inside of the vehicle to the outside of the vehicle, thus achieving the effect of thermal insulation in summer and heat retention in winter. Meanwhile, by controlling the T_{nIR} of the laminated glass 10 for infrared with a wavelength from 1000 nm to 1250 nm and the R_{IR} of the laminated glass 10 for infrared with a wavelength from 780 nm to 2500 nm, the range of the thermal insulation index C of the laminated glass 10 may be $0.008 \leq C \leq 0.4$, so that the laminated glass 10 has an ideal thermal blocking effect, which may provide a good sensation on skins of passengers. When the laminated glass 10 is mounted on the vehicle, an ideal thermal blocking effect may be achieved without a sunshade or other accessories, thereby meeting requirements such as lightweight of vehicles, increasing headroom, reducing costs.

[0025] Specifically, the value of thermal insulation index of the laminated glass 10 may be, but is not limited to 0.008, 0.009, 0.01, 0.02, 0.05, 0.06, 0.1, 0.13, 0.15, 0.2, 0.24, 0.27, 0.31, 0.35, 0.37, 0.4, etc.

[0026] Optionally, in some embodiments, there is one transparent conductive oxide coating 131 in the low-emissivity coating 13. In some other embodiments, there are two transparent conductive oxide coatings 131 in the low-emissivity coating 13. In some other embodiments, there are three transparent conductive oxide coatings 131 in the low-emissivity coating 13.

[0027] In embodiments of the disclosure, a surface of the outer glass plate 15 away from the adhesive layer 17 is the first surface 151, and a surface of the outer glass plate 15 close to the adhesive layer 17 is the second surface 152. A surface of the inner glass plate 11 away from the adhesive layer 17 is the third surface 111, and a surface of the inner glass plate 11 close to the adhesive layer 17 is the fourth surface 112. When the laminated glass 10 is applied to the vehicle 30, the first surface 151 is the outermost surface of the laminated glass 10 close to the outside of the vehicle, and the fourth surface 112 is the innermost surface of the laminated glass 10 close to the inside of the vehicle.

[0028] The laminated glass 10 in the disclosure may be applied to a building, the vehicle 30, etc. When the laminated glass 10 is applied to the vehicle 30, the laminated glass 10 may be applied to at least one of a sunroof glass, a side window glass, or a rear window glass of the vehicle 30.

[0029] Human skin is most sensitive to heat with a wavelength of 1000 nm to 1250 nm. Therefore, by controlling the T_{nIR} of the laminated glass for infrared with a wavelength of 1000 nm to 1250 nm and the R_{IR} of the laminated glass for infrared with a wavelength of 780 nm to 2500 nm, the thermal insulation index C may be controlled within a certain range. In this way, the laminated glass 10 not only blocks the transmission of heat from the outside of the vehicle 30 to the interior space of the vehicle 30 effectively,

but also blocks the dissipation of heat from the interior space of the vehicle 30 to the outside of the vehicle 30, and the laminated glass 10 has a better low-emissivity property. Meanwhile, the laminated glass 10 has an ideal thermal blocking effect, which may provide a good sensation on skins of passengers. When the laminated glass 10 is mounted on the vehicle, an ideal thermal blocking effect may be achieved without a sunshade or other accessories, thereby meeting requirements such as lightweight of vehicles, increasing headroom, reducing costs. When the thermal insulation index C satisfies $0.008 \leq C \leq 0.4$, the laminated glass 10 has an ideal thermal blocking effect, which may provide a good sensation on skins of passengers. When the thermal insulation index C is greater than 0.4, the laminated glass 10 has a high transmittance for infrared with a wavelength of 1000 nm to 1250 nm and has a low reflectivity for infrared with a wavelength of 780 nm to 2500 nm, resulting in a poor thermal blocking effect of the laminated glass and a poor sensation on skins of passengers. When the thermal insulation index C is less than 0.008, the laminated glass 10 has an excellent thermal blocking effect. However, there is a higher requirement for the production materials and processes for making the laminated glass 10, which will further increase the production cost of the laminated glass 10.

[0030] Optionally, in some embodiments, the range of the thermal insulation index C of the laminated glass 10 is $0.03 \leq C \leq 0.25$. Specifically, the thermal insulation index C of the laminated glass 10 may be, but is not limited to, 0.03, 0.05, 0.08, 0.12, 0.13, 0.15, 0.18, 0.19, 0.20, 0.22, 0.24, 0.25, etc. When the thermal insulation index C of the laminated glass 10 satisfies $0.03 \leq C \leq 0.25$, the laminated glass 10 has an ideal thermal blocking effect, which may provide a good sensation on skins of passengers.

[0031] Optionally, in some other embodiments, the range of the thermal insulation index C of the laminated glass 10 is $0.009 \leq C \leq 0.15$. Specifically, the thermal insulation index C of the laminated glass 10 may be, but is not limited to, 0.009, 0.01, 0.03, 0.05, 0.07, 0.09, 0.10, 0.11, 0.22, 0.14, 0.15, etc. When the thermal insulation index C of the laminated glass 10 satisfies $0.009 \leq C \leq 0.15$, the laminated glass 10 has an ideal thermal blocking effect, which may provide a good sensation on skins of passengers.

[0032] In an embodiment of the disclosure, the visible light transmittance of the laminated glass 10 ranges from 0.5% to 10%. Specifically, the visible light transmittance of the laminated glass 10 may be, but is not limited to, 0.5%, 1.5%, 2%, 2.5%, 3.3%, 4.8%, 5%, 6.7%, 8%, 9.4%, 10%, etc.

[0033] When the visible light transmittance of the laminated glass 10 ranges from 0.5% to 10%, the laminated glass 10 has a low visible light transmittance, which may protect the privacy of passengers and serve as a substitute for the sunshade, making it suitable for being applied to the sunroof glass, the side window glass, and the rear window glass of the vehicle 30.

[0034] In an embodiment of the disclosure, the emissivity of the laminated glass 10 measured at the side of the low-emissivity coating 13 away from the inner glass plate 11 is less than or equal to 0.25. Specifically, the emissivity of the laminated glass 10 may be, but is not limited to 0.12, 0.14, 0.17, 0.19, 0.21, 0.22, 0.23, 0.25, etc.

[0035] When the emissivity of the laminated glass 10 is smaller than or equal to 0.25, the laminated glass 10 has a low-emissivity property, so that the laminated glass 10 can

effectively block the transmission of heat from the outside of the vehicle 30 to the interior space of the vehicle in summer and block the dissipation of heat from the interior space of the vehicle 30 to the outside of the vehicle 30 in winter. Therefore, the effect of heat insulation in summer and heat retention in winter may be achieved.

[0036] In an embodiment of the disclosure, the visible light reflectivity of the laminated glass 10 measured at the side of the low-emissivity coating 13 away from the inner glass plate 11 is less than or equal to 4%. Specifically, the reflectivity of the laminated glass 10 may be, but is not limited to, 4%, 3.8%, 3.75%, 3.6%, 3.5%, 3.2%, 3.1%, 3.0%, etc.

[0037] The visible light reflectivity of the laminated glass 10 measured at the side of the low-emissivity coating 13 away from the inner glass plate 11 is low. When the laminated glass 10 is applied to the vehicle 30, especially to the sunroof glass of the vehicle 30, the laminated glass 10 can reduce or even eliminate the formation of obvious reflections of passengers and objects on the sunroof glass due to specular reflection, thereby reducing visual interference caused by reflections for passengers, especially those in the rear seats, and improving the passenger experience.

[0038] In an embodiment, the visible light reflectivity of the laminated glass 10 is less than or equal to 3%. Specifically, the visible light reflectivity of the laminated glass 10 may be, but is not limited to, 3%, 2.8%, 2.75%, 2.6%, 2.5%, 2.2%, 2.1%, 2.0%, etc.

[0039] In an embodiment, the visible light reflectivity of the laminated glass 10 is less than or equal to 2%. Specifically, the visible light reflectivity of the laminated glass 10 may be, but is not limited to, 2%, 1.8%, 1.75%, 1.6%, 1.5%, 1.2%, 1.1%, 0.9%, 0.8%, 0.6%, etc.

[0040] In an embodiment of the disclosure, the inner glass plate 11 is a tinted glass.

[0041] It may be understood that, the inner glass plate 11 is a tinted glass. Optionally, the inner glass plate 11 may be, but is not limited to any one of a green glass, a grey glass, a blue glass, or a brown glass.

[0042] In some embodiments of the disclosure, the visible light transmittance of the inner glass plate 11 ranges from 5% to 45%. Specifically, the visible light transmittance of the inner glass plate 11 may be, but is not limited to, 5%, 8%, 16%, 19%, 23%, 25%, 33%, 35%, 37%, 40%, etc.

[0043] Optionally, in some embodiments, when the visible light transmittance of the inner glass plate 11 ranges from 34% to 42%, a value range of L in the Lab value of the color of the inner glass plate 11 is 65 to 71.5, a value range of a is -3.5 to -2 , and a value range of b is 2 to 3.5. In the embodiment, the inner glass plate 11 is a grey glass.

[0044] Specifically, the visible light transmittance of the inner glass plate 11 may be, but is not limited to, 34%, 35%, 36%, 37%, 39%, 40%, 41%, 42%, etc. In the Lab value of the color of the inner glass plate 11, the value of L may be, but is not limited to, 65, 65.3, 66.4, 66.6, 67.2, 67.9, 68.5, 69.3, 69.7, 70.2, 70.9, 71.5, etc. The value of a may be, but is not limited to, -3.5 , -3.4 , -3.1 , -2.9 , -2.7 , -2.5 , -2.3 , -2.1 , -2 , etc. The value of b may be, but is not limited to, 2, 2.1, 2.3, 2.4, 2.7, 3.0, 3.1, 3.3, 3.4, 3.5, etc.

[0045] Optionally, in some other embodiments, when the visible light transmittance of the inner glass plate 11 ranges from 26% to 34%, a value range of Z in the Lab value of the color of the inner glass plate 11 is 58.5 to 65, a value range

of a is -5 to -3.5 , and a value range of b is 0.5 to 2 . In the embodiment, the inner glass plate **11** is a grey glass.

[0046] Specifically, the visible light transmittance of the inner glass plate **11** may be, but is not limited to, 26%, 26.3%, 27.1%, 27.7%, 28.4%, 29%, 30.4%, 31.4%, 32.2%, 32.8%, 33.6%, 34%, etc. In the Lab value of the color of the inner glass plate **11**, the value of L may be, but is not limited to, 58.5, 58.9, 59.4, 59.8, 60.2, 60.8, 61.4, 61.8, 62.7, 63.5, 64.2, 64.8, 65, etc. The value of a may be, but is not limited to, -5 , -4.9 , -4.7 , -4.5 , -4.4 , -4.2 , -4.0 , -3.9 , -3.8 , -3.7 , -3.6 , -3.5 , etc. The value of b may be, but is not limited to, 0.5 , 0.6 , 0.8 , 1.0 , 1.2 , 1.3 , 1.5 , 1.7 , 1.8 , 1.9 , 2 , etc.

[0047] In an embodiment of the disclosure, the mass fraction of total iron as Fe_2O_3 in the inner glass plate **11** ranges from 0.9% to 2.2% . Specifically, the mass fraction of total iron as Fe_2O_3 in the inner glass plate **11** may be, but is not limited to 0.9% , 1.0% , 1.1% , 1.3% , 1.4% , 1.7% , 1.8% , 2.0% , 2.1% , 2.2% , etc.

[0048] When the mass fraction of total iron as Fe_2O_3 in the inner glass plate **11** ranges from 0.9% to 2.2% , the T_{NIR} of the laminated glass **10** for infrared with a wavelength of 1000 nm to 1250 nm is within a reasonable range, so that the laminated glass **10** can effectively block the infrared with a wavelength of 1000 nm to 1250 nm from entering the interior space of the vehicle **30** through the laminated glass **10**. In this way, the laminated glass **10** has an ideal thermal blocking effect, which may provide a good sensation on skins of passengers.

[0049] The iron oxides in the glass mainly serve as coloring components and functional components. The iron in the glass predominantly exists in the form of Fe^{2+} and Fe^{3+} . Fe^{2+} has an absorption peak near the wavelength of 1100 nm , and Fe^{3+} has an absorption peak near the wavelength of 400 nm . By adjusting the mass of total iron in the inner glass plate **11**, the visible light transmittance and the T_{NIR} of the laminated glass **10** may be adjusted, particularly the T_{NIR} for infrared with a wavelength near 1100 nm . The total iron in the disclosure refers to all iron oxides in the inner glass plate **11**, including both Fe^{2+} (FeO) and Fe^{3+} (Fe_2O_3), which is a common expression in the related field and does not imply that all iron oxides in the glass are Fe_2O_3 .

[0050] Optionally, the thickness of the inner glass plate **11** ranges from 0.7 mm to 2.1 mm . Specifically, the thickness of the inner glass plate **11** may be, but is not limited to, 0.7 mm , 0.9 mm , 1.0 mm , 1.1 mm , 1.3 mm , 1.5 mm , 1.7 mm , 1.8 mm , 2.0 mm , 2.1 mm , etc.

[0051] When the thickness of the inner glass plate **11** ranges from 0.7 mm to 2.1 mm , the thickness of the inner glass plate **11** is within a reasonable range, so that the inner glass plate **11** may be applied to the vehicle **30** after being combined with the outer glass plate **15** by the adhesive layer **17**.

[0052] Optionally, in terms of mass fraction, the inner glass plate **11** includes 1.1% to 1.9% of total iron as Fe_2O_3 , 140 part per million (ppm) to 300 ppm of Co_2O_3 , 10 ppm to 320 ppm of Cr_2O_3 , 55 ppm to 75 ppm of SrO , and 10 ppm to 30 ppm of Se .

[0053] Specifically, the mass fraction of total iron as Fe_2O_3 in the inner glass plate **11** may be, but is not limited to, 1.1% , 1.2% , 1.3% , 1.4% , 1.5% , 1.6% , 1.7% , 1.8% , and 1.9% . The mass fraction of Co_2O_3 in the inner glass plate **11** may be, but is not limited to, 140 ppm , 146 ppm , 152 ppm , 165 ppm , 170 ppm , 185 ppm , 196 ppm , 206 ppm , 218 ppm , 225 ppm , 240 ppm , 250 ppm , 260 ppm , 270 ppm , 280 ppm ,

300 ppm , etc. The mass fraction of Cr_2O_3 in the inner glass plate **11** may be, but is not limited to, 10 ppm , 16 ppm , 25 ppm , 38 ppm , 56 ppm , 88 ppm , 107 ppm , 138 ppm , 156 ppm , 188 ppm , 220 ppm , 240 ppm , 270 ppm , 298 ppm , 300 ppm , 320 ppm , etc. The mass fraction of SrO in the inner glass plate **11** may be, but is not limited to, 55 ppm , 56 ppm , 58 ppm , 62 ppm , 65 ppm , 68 ppm , 71 ppm , 72 ppm , 73 ppm , 75 ppm , etc. The mass fraction of Se in the inner glass plate **11** may be, but is not limited to, 10 ppm , 11 ppm , 15 ppm , 18 ppm , 21 ppm , 22 ppm , 23 ppm , 25 ppm , 28 ppm , 30 ppm , etc. In the embodiments of the disclosure, the inner glass plate **11** is a tinted glass, which can block infrared from passing through the inner glass plate **11**, so that the laminated glass **10** has a good heat insulation effect and a suitable visible light transmittance.

[0054] Optionally, the material of the transparent conductive oxide coating **131** is selected from at least one of doped zinc oxide (ZnO), indium tin oxide (ITO), nickel chromium oxide (CrNiO_x), fluorine-doped tin oxide (FTO), or zinc tin oxide (ZnSnO_x), where the doped zinc oxide is zinc oxide doped with one or more elements selected from a group consisting of Al, tungsten (W), hafnium (Hf), gallium (Ga), yttrium (Y), niobium (Nb), and neodymium (Nd).

[0055] Optionally, the low-emissivity coating **13** may be directly deposited onto the fourth surface **112** through chemical vapor deposition (CVD) or physical vapor deposition (PVD). The total thickness of the transparent conductive oxide coating **131** in the low-emissivity coating **13** ranges from 50 nm to 300 nm . In the case where the low-emissivity coating **13** is not provided, the emissivity of the laminated glass **10** is about 0.9 . The low-emissivity coating **13** can reduce the emissivity of the laminated glass **10**, so that the emissivity of the laminated glass **10** may be less than or equal to 0.25 . Emissivity is primarily used to measure the ability of an object surface to release energy in the form of radiation, usually reflecting the ability to block mid-to-far infrared (wavelengths greater than 2500 nm).

[0056] Optionally, in an embodiment of the disclosure, the transparent conductive oxide coating **131** is an ITO layer deposited through magnetron sputtering process. The target material of the transparent conductive oxide coating **131** includes indium oxide and tin oxide, with a ratio of the mass of indium oxide to the mass of tin oxide ranging from $80:20$ to $90:10$. Specifically, the ratio of the mass of indium oxide to the mass of tin oxide may be, but is not limited to, $80:20$, $82:18$, $85:15$, $88:12$, $90:10$, etc.

[0057] Optionally, the total thickness of the transparent conductive oxide coating **131** in the low-emissivity coating **13** ranges from 50 nm to 300 nm . Specifically, the total thickness of the transparent conductive oxide coating **131** may be, but is not limited to, 50 nm , 70 nm , 90 nm , 130 nm , 180 nm , 190 nm , 220 nm , 230 nm , 260 nm , 300 nm , etc.

[0058] When the total thickness of the transparent conductive oxide coating **131** ranges from 50 nm to 300 nm , the laminated glass **10** has a low emissivity. When the total thickness of the transparent conductive oxide coating **131** is less than 50 nm , the laminated glass **10** has a poor low-emissivity property and cannot effectively block the transmission of heat from the inside of the vehicle **30** to the outside or the transmission of heat from the outside of the vehicle **30** to the inside. When the total thickness of the transparent conductive oxide coating **131** is greater than 300 nm , the visible light reflectivity of the laminated glass **10**

also increases, which reduces the anti-reflection effect of the laminated glass 10, causing interference and discomfort to passengers.

[0059] In an embodiment of the disclosure, the transparent conductive oxide coating 131 is an ITO layer, the inner glass plate 11 includes iron oxide, and the ratio of the mass of total iron as Fe_2O_3 in the inner glass plate 11 to the mass of indium in the 1 to 3 transparent conductive oxide coatings 131 ranges from 20:1 to 500:1, preferably from 50:1 to 200:1. Specifically, the ratio of the mass of total iron as Fe_2O_3 in the inner glass plate 11 to the mass of indium in the 1 to 3 transparent conductive oxide coatings 131 may be, but is not limited to, 20:1, 30:1, 50:1, 80:1, 100:1, 110:1, 120:1, 150:1, 180:1, 200:1, 250:1, 300:1, 350:1, 400:1, 450:1, 500:1, etc.

[0060] When the transparent conductive oxide coating 131 is an ITO layer deposited through magnetron sputtering process, the low-emissivity coating 13 including the ITO layer is disposed on the fourth surface 112 of the inner glass plate 11. Increasing the mass of total iron as Fe_2O_3 in the inner glass plate 11 can reduce the visible light transmittance of the laminated glass 10 and also decrease the T_{IR} for infrared with a wavelength of 1000 nm to 1250 nm. Correspondingly, increasing the mass of total iron as Fe_2O_3 in the inner glass plate 11 increases the manufacturing difficulty and cost of the inner glass plate 11, and results in that the laminated glass 10 is easier to absorb heat to heat up in summer. Increasing the mass of indium in the 1 to 3 transparent conductive oxide coatings 131 can reduce the emissivity of the laminated glass 10, and increase the reflectivity of the laminated glass 10 for infrared with a wavelength from 780 nm to 2500 nm. Correspondingly, increasing the mass of indium in the 1 to 3 transparent conductive oxide coatings 131 also increases the manufacturing cost and difficulty of the laminated glass 10, and results in that the low-emissivity coating 13 may not meet the automotive-grade requirements for vehicle glass. In this embodiment of the disclosure, when the ratio of the mass of total iron as Fe_2O_3 in the inner glass plate 11 to the mass of indium in the 1 to 3 transparent conductive oxide coatings 131 ranges from 20:1 to 500:1, it is easier to achieve that the thermal insulation index C of the laminated glass 10 satisfies $0.008 \leq C \leq 0.4$, or even $0.03 \leq C \leq 0.25$, or $0.009 \leq C \leq 0.15$, resulting in an ideal thermal blocking effect and providing a good sensation on skins of passengers. When the ratio of the mass of total iron as Fe_2O_3 in the inner glass plate 11 to the mass of indium in the 1 to 3 transparent conductive oxide coatings 131 is greater than 500:1, the inner glass plate 11 is easier to absorb heat and radiate more heat to the inside of the vehicle in summer, resulting in a poor thermal blocking effect and a poor sensation on skins of passengers. Also, it becomes more difficult to coordinate the mass of total iron with other components, thereby raising the manufacturing cost and difficulty of the inner glass plate 11. When the ratio of the mass of total iron as Fe_2O_3 in the inner glass plate 11 to the mass of indium in the 1 to 3 transparent conductive oxide coatings 131 is less than 20:1, the total thickness of the ITO layer is too large, resulting in higher requirements for material and process of the low-emissivity coating 13. The produced low-emissivity coating 13 may not meet the automotive-grade requirements for vehicle glass, thereby further increasing the manufacturing cost of the laminated glass 10.

[0061] Reference is made to FIG. 3. In an embodiment of the disclosure, the low-emissivity coating 13 further includes at least two first dielectric layers 132, a first dielectric layer 132 is disposed on each of two opposite surfaces of each transparent conductive oxide coating 131. In the preparation process of the laminated glass 10, high-temperature heat treatment and bending processes are required. The first dielectric layer 132 can protect the transparent conductive oxide coating 131 from corrosion or damage during the preparation process or usage, can adjust the optical effect and appearance color of the low-emissivity coating 13, and can improve the adhesion performance between the low-emissivity coating 13 and the inner glass plate 11, thereby being conducive to achieving the thermal blocking effect of the laminated glass 10. In addition, reasonable design and combination of the material and the thickness of the first dielectric layer 132 are beneficial for achieving an antireflective effect of the low-emissivity coating 13, which can reduce or even eliminate the formation of obvious reflections of passengers and objects on the sunroof glass due to specular reflection, thereby reducing visual interference caused by reflections for passengers, especially those in the rear seats, and improving the passenger experience.

[0062] Optionally, the material of the first dielectric layer 132 is selected from nitride, oxide, or oxynitride of at least one of Zn, Sn, Ti, Si, Al, Mg, or Zr. Specifically, the material of the first dielectric layer 132 may include, but is not limited to, zinc nitride, zinc oxide, zinc oxynitride, titanium oxynitride, magnesium oxide, aluminum nitride, etc.

[0063] Optionally, the thickness of the first dielectric layer 132 ranges from 5 nm to 100 nm. Specifically, the thickness of the first dielectric layer 132 may be, but is not limited to, 5 nm, 10 nm, 25 nm, 38 nm, 45 nm, 50 nm, 68 nm, 75 nm, 80 nm, 100 nm, etc.

[0064] In embodiments of the disclosure, when the thickness of the first dielectric layer 132 ranges from 5 nm to 100 nm, the first dielectric layer 132 can protect the transparent conductive oxide coating 131 from corrosion or damage during the preparation process or usage, can adjust the optical effect and appearance color of the low-emissivity coating 13, and can improve the adhesion performance between the low-emissivity coating 13 and the inner glass plate 11, thereby being conducive to achieving the thermal blocking effect of the laminated glass 10.

[0065] When the thickness of the first dielectric layer 132 is greater than 100 nm, the first dielectric layer 132 is too thick, making it difficult for the first dielectric layer 132 to match with other coatings, thus reducing the thermal blocking effect of the laminated glass 10. When the thickness of the first dielectric layer 132 is less than 5 nm, the first dielectric layer 132 is too thin and may be easily worn-out during actual use, making it difficult for the first dielectric layer 132 to continue protecting the transparent conductive oxide coating 131 from corrosion or damage.

[0066] Reference is made to FIG. 4 and FIG. 5. In some embodiments, the laminated glass 10 further includes an infrared reflection coating 18. The infrared reflection coating 18 is disposed between the inner glass plate 11 and the outer glass plate 15, the infrared reflection coating 18 includes 1 to 5 metal layers 184, and a material of the metal layer 184 is selected from metal or alloy of at least one of Ag, Au, Cu, Al, or Pt. Specifically, the infrared reflection coating 18 may be directly disposed on the second surface 152 of the outer

glass plate **15** or on the third surface **111** of the inner glass plate **11**. The infrared reflection coating **18** may also be firstly disposed on an organic resin film, such as polyethylene glycol terephthalate, and then the organic resin film with the infrared reflection coating **18** is interposed between the second surface **152** and the third surface **111**. In this way, the laminated glass **10** may have a visible light transmittance of less than or equal to 10% and a total solar transmittance (TTS) of less than or equal to 25%.

[0067] Optionally, the thickness of the infrared reflection coating **18** ranges from 40 nm to 500 nm. Specifically, the thickness of the infrared reflection coating **18** may be, but is not limited to 40 nm, 50 nm, 80 nm, 140 nm, 160 nm, 220 nm, 280 nm, 330 nm, 380 nm, 440 nm, 480 nm, 500 nm, etc.

[0068] When the thickness of the infrared reflection coating **18** ranges from 40 nm to 500 nm, the infrared reflection coating **18** can effectively reflect infrared in sunlight, thereby blocking the transmission of heat, improving the thermal blocking ability of the laminated glass **10**, and reducing the TTS of the laminated glass **10**.

[0069] Optionally, the material of the metal layer **184** may be, but is not limited to, silver (Ag), aurum (Au), copper (Cu), aluminum (Al), platinum (Pt), silver-copper alloy, copper-aluminum alloy, copper-gold alloy, etc. Among those materials, silver or silver alloys are commonly used, and the silver alloys may be silver-gold alloy, silver-aluminum alloy, silver-copper alloy, silver-platinum alloy, etc.

[0070] Reference is made to FIG. 6. In some embodiments, in a case where there is only one metal layer **184** in the infrared reflection coating **18**, the infrared reflection coating **18** may be called as a single-silver infrared reflection coating. Similarly, when the infrared reflection coating **18** includes two metal layers **184**, the infrared reflection coating **18** may be called as a double-silver infrared reflection coating. Similarly, there may be a triple-silver infrared reflection coating, a quadruple-silver infrared reflection coating, or a quintuple-silver infrared reflection coating.

[0071] Optionally, the thickness of each metal layer **184** in the infrared reflection coating **18** ranges from 4 nm to 20 nm. Specifically, the thickness of each metal layer **184** may be, but is not limited to, 4 nm, 5 nm, 6 nm, 8 nm, 10 nm, 12 nm, 14 nm, 16 nm, 17 nm, 19 nm, 20 nm, etc.

[0072] When the thickness of each metal layer **184** ranges from 4 nm to 20 nm, it allows the infrared reflection coating **18** to have a good infrared reflection effect, and it is beneficial for reducing the design difficulty of the infrared reflection coating **18** and ensuring that the infrared reflection coating **18** can undergo high-temperature heat treatment and bending processes, thus meeting the automotive-grade requirements for the vehicle window.

[0073] Reference is made to FIG. 6 and FIG. 7. In an embodiment of the disclosure, the infrared reflection coating **18** further includes at least two second dielectric layers **181**, and a second dielectric layer **181** is disposed on each of two opposite surfaces of each metal layer **184**. In the preparation process of the laminated glass **10**, high-temperature heat treatment and bending processes are required. The second dielectric layer **181** can protect the metal layer **184** from oxidation or damage during the preparation process or usage, and can adjust the optical effect and appearance color of the infrared reflection coating **18**, thereby improving the adhesion performance of the infrared reflection coating **18**, which is conducive to achieving the thermal blocking effect of the laminated glass **10**.

[0074] Reference is made to FIG. 6 and FIG. 7. The infrared reflection coating **18** is disposed on the second surface **152** of the outer glass plate **15**. For the second dielectric layers **181** disposed on the metal layer **184**, one that is closer to the second surface **152** is a second inner dielectric layer **182**, and the other that is farther from the second surface **152** is the second outer dielectric layer **183**. When the infrared reflection coating **18** is disposed on the third surface **111** of the inner glass plate **11**, for the second dielectric layers **181** disposed on the metal layer **184**, one that is closer to the third surface **111** is the second inner dielectric layer **182**, and the other that is farther from the third surface **111** is the second outer dielectric layer **183**. It may be understood that, the total number of the second dielectric layers **181** is one more than the total number of the metal layers **184**. For example, the infrared reflection coating **18** includes one metal layer **184** and two second dielectric layers **181**, or two metal layers **184** and three second dielectric layers **181**, or three metal layers **184** and four second dielectric layers **181**, or four metal layers **184** and five second dielectric layers **181**, or five metal layers **184** and six second dielectric layers **181**.

[0075] The material of the second dielectric layer is selected from nitride, oxide, or oxynitride of at least one element of Zn, Sn, Ti, Si, Al, Ni, Cr, Nb, Mg, Zr, Ga, Y, In, Sb, V, or Ta. Optionally, the material of the second dielectric layer may be, but is not limited to, zinc stannate (ZnSnO_3), magnesium-doped zinc stannate, zinc oxide, magnesium-doped zinc oxide, zirconium-doped zinc oxide, niobium oxide, bismuth oxide, aluminum-doped zinc oxide (AZO), zirconium oxide, titanium oxide, peroxy titanium, nickel-chromium, zirconium nitride, silicon nitride, silicon oxide, silicon oxynitride, etc.

[0076] In embodiments of the disclosure, the laminated glass **10** includes the low-emissivity coating **13**, the inner glass plate **11**, the adhesive layer **17**, the infrared reflection coating **18**, and the outer glass plate **15** stacked in sequence in a direction from the inside to the outside of the vehicle **30**. The infrared reflection coating **18** can reflect infrared in sunlight, thereby improving the thermal blocking effect of the laminated glass **10** and effectively blocking the transmission of heat from the exterior to the interior space of the vehicle **30**. When the laminated glass **10** is applied to the vehicle **30**, an ideal thermal blocking effect may be achieved without a sunshade or other accessories, thereby meeting the requirements such as lightweight of vehicles and reducing costs.

[0077] In some embodiments, the outer glass plate **15** is a clear glass with a visible light transmittance greater than or equal to 80%. When the laminated glass **10** includes the infrared reflection coating **18**, the laminated glass **10** being a clear glass can maximize the ability of the infrared reflection coating **18** of reflecting infrared in sunlight, thereby reflecting as many infrareds in sunlight as possible and absorbing as few infrareds in sunlight as possible, so that an ideal thermal blocking effect may be achieved. Specifically, the visible light transmittance of the outer glass plate **15** may be, but is not limited to, 80%, 82%, 84%, 85%, 88%, etc. In an embodiment, the visible light transmittance of the outer glass plate **15** is greater than or equal to 90%, and specifically, the visible light transmittance of the outer glass plate **15** may be, but is not limited to, 90%, 92%, 94%, 95%, 98%, etc.

[0078] Optionally, in some embodiments, when the laminated glass **10** does not include the infrared reflection coating **18**, the outer glass plate **15** is a tinted glass, and the mass fraction of total iron as Fe_2O_3 in the outer glass plate **15** ranges from 0.7% to 2.2%. Specifically, the mass fraction of total iron as Fe_2O_3 in the outer glass plate **15** may be, but is not limited to, 0.7%, 0.8%, 0.9%, 1%, 1.1%, 1.2%, 1.4%, 1.6%, 1.8%, 1.9%, 2.0%, 2.1%, 2.2%, etc.

[0079] When the mass fraction of total iron as Fe_2O_3 in the outer glass plate **15** ranges from 0.7% to 2.2%, the T_{IR} of the laminated glass **10** for infrared with a wavelength of 1000 nm to 1250 nm is within a reasonable range, such that the laminated glass **10** can effectively block the infrared with a wavelength of 1000 nm to 1250 nm from passing through the laminated glass **10** into the interior space of the vehicle **30**. In this way, the laminated glass **10** has an ideal thermal blocking effect, which provides a good sensation on skins of passengers. It may further ensure that the laminated glass **10** has a visible light transmittance of less than or equal to 10% and a TTS of less than or equal to 25%.

[0080] In embodiments of the disclosure, the thickness of the outer glass plate **15** ranges from 2.1 mm to 4.2 mm. Specifically, the thickness of the outer glass plate **15** may be, but is not limited to, 2.1 mm, 2.3 mm, 2.5 mm, 2.7 mm, 2.9 mm, 3.0 mm, 3.2 mm, 3.5 mm, 3.85 mm, 4.0 mm, 4.2 mm, etc.

[0081] Optionally, the outer glass plate **15** may be, but is not limited to any one of a green glass, a grey glass, a blue glass, or a brown glass.

[0082] In some embodiments of the disclosure, the visible light transmittance of the outer glass plate **15** ranges from 5% to 85%. Specifically, the visible light transmittance of the outer glass plate **15** may be, but is not limited to, 5%, 8%, 16%, 19%, 23%, 25%, 33%, 35%, 37%, 40%, 50%, 60%, 70%, 82%, etc. When the visible light transmittance of the outer glass plate **15** ranges from 5% to 85%, the difficulty of high-temperature heat treatment and bending processes for the laminated glass may be reduced.

[0083] Optionally, the adhesive layer **17** is disposed between the inner glass plate **11** and the outer glass plate **15**, and the adhesive layer **17** is used to bond the inner glass plate **11** and the outer glass plate **15**. In some embodiments, the adhesive layer **17** is a transparent thermoplastic polymer film with a visible light transmittance of greater than or equal to 70%. When the adhesive layer **17** is a transparent thermoplastic polymer film with a visible light transmittance of greater than or equal to 70%, the laminated glass **10** has a higher visibility, which allows passengers to have a clearer view when the laminated glass **10** is applied to a vehicle **30**. Thus, the laminated glass **10** may be used in side window glass, rear windshield glass, etc., of the vehicle **30**. Specifically, the visible light transmittance of the adhesive layer **17** may be, but is not limited to, 70%, 71%, 73%, 75%, 78%, 80%, 82%, etc. In other embodiments, the adhesive layer **17** is a tinted thermoplastic polymer film with a visible light transmittance ranging from 1% to 20%. In this case, the adhesive layer **17** reduces the visible light transmittance of the laminated glass **10**, but improves the infrared-blocking ability of the laminated glass **10**, thereby providing a good sensation on skins of passengers. However, the adhesive layer **17** provided in the disclosure has a lower visible light transmittance, resulting in that the laminated glass **10** has a lower visible light transmittance. Therefore, the laminated glass **10** is suitable for use only in sunroof glass, which may

prevent passengers in the vehicle from being exposed to the glare of sunlight and provide heat insulation. Specifically, the visible light transmittance of the adhesive layer **17** may be, but is not limited to, 1%, 2%, 5%, 7%, 8%, 10%, 12%, 14%, 15%, 18%, 20%, etc.

[0084] Optionally, the thermoplastic polymer film is selected from at least one of polyvinyl butyral (PVB) film, polyurethane (PU) film, ethylene-vinyl acetate copolymer (EVA) film, or ionic polymer (e.g., SentryGlas® Plus (SGP)) film. The thermoplastic polymer film can bond the inner glass plate **11** and the outer glass plate **15**, and has good adhesion performance, which can effectively improve the bonding strength between the inner glass plate **11** and the outer glass plate **15**.

[0085] Reference is made to FIG. 8 to FIG. 10. In an embodiment of the disclosure, the laminated glass **10** further includes a light-adjustment component **19** disposed between the inner glass plate **11** and the outer glass plate **15**. The light-adjustment component **19** is configured to adjust the visible light transmittance of the laminated glass **10** to meet the optical performance requirements of different usage scenarios.

[0086] It may be understood that, when the laminated glass **10** includes the light-adjustment component **19**, the number of adhesive layers **17** is 2, and the light-adjustment component **19** is disposed between two adjacent adhesive layers **17**.

[0087] Specifically, in some embodiments, the laminated glass **10** includes the low-emissivity coating **13**, the inner glass plate **11**, the adhesive layer **17**, the light-adjustment component **19**, the adhesive layer **17**, and the outer glass plate **15** stacked in sequence. In other embodiments, the laminated glass **10** includes the low-emissivity coating **13**, the inner glass plate **11**, the adhesive layer **17**, the light-adjustment component **19**, the adhesive layer **17**, the infrared reflection coating **18**, and the outer glass plate **15** stacked in sequence. In other embodiments, the laminated glass **10** includes the low-emissivity coating **13**, the inner glass plate **11**, the adhesive layer **17**, the light-adjustment component **19**, the adhesive layer **17**, the infrared reflection coating **18**, an organic resin film (polyethylene terephthalate, PET), the adhesive layer **17**, and the outer glass plate **15** stacked in sequence.

[0088] Optionally, the light-adjustment component **19** includes at least one of polymer dispersed liquid crystal (PDLC) light-adjustment film, suspended particle device (SPD) light-adjustment film, or electrochromic (EC) light-adjustment film.

[0089] The laminated glass **10** provided in the disclosure will be further described through specific embodiments in the following.

Examples 1 to 5

[0090] In examples 1 to 5, the laminated glass **10** each includes the low-emissivity coating **13**, the inner glass plate **11**, the adhesive layer **17**, and the outer glass plate **15** stacked in sequence. The low-emissivity coating **13** includes a Si_3N_4 layer, a SiO_2 layer, an ITO layer, a SiO_2 layer, an ITO layer, a Si_3N_4 layer, and a SiO_2 layer stacked in sequence on the fourth surface **112** of the inner glass plate **11** through magnetron sputtering process, where the ITO layer is the transparent conductive oxide coating and the others are the first dielectric layers. The low-emissivity

coating **13** includes two ITO layers, and the target material for magnetron sputtering the ITO layer has a mass ratio of In_2O_3 to SnO_2 of 90:10.

Example 1

[0091] The inner glass plate **11** is a grey glass with the thickness of 2.1 mm and has a visible light transmittance of 28%. The adhesive layer **17** is a grey PVB film with the thickness of 0.76 mm and has a visible light transmittance of 2%. The outer glass plate **15** is a green glass with the thickness of 2.1 mm and has a visible light transmittance of 82%. The low-emissivity coating **13** includes a Si_3N_4 layer (6 nm), a SiO_2 layer (63 nm), an ITO layer (20 nm), a SiO_2 layer (34 nm), an ITO layer (120 nm), a Si_3N_4 layer (18 nm), and a SiO_2 layer (82 nm) stacked in sequence in a direction away from the fourth surface **112**.

Example 2

[0092] The configuration of example 2 is the same as in example 1, except that the adhesive layer **17** is a grey PVB film with the thickness of 0.76 mm and has a visible light transmittance of 8%.

Example 3

[0093] The configuration of example 3 is the same as in example 1, except that the outer glass plate **15** is a grey glass with the thickness of 2.1 mm and has a visible light transmittance of 40%, and the adhesive layer **17** is a grey PVB film with the thickness of 0.76 mm and has a visible light transmittance of 8%.

Example 4

[0094] The configuration of example 4 is the same as in example 1, except that the outer glass plate **15** is a grey glass with the thickness of 2.1 mm and has a visible light transmittance of 40%, the adhesive layer **17** is a grey PVB film with the thickness of 0.76 mm and has a visible light transmittance of 18%, and the low-emissivity coating **13** includes a Si_3N_4 layer (6 nm), a SiO_2 layer (63 nm), an ITO layer (20 nm), a SiO_2 layer (34 nm), an ITO layer (80 nm), a Si_3N_4 layer (18 nm), and a SiO_2 layer (82 nm) stacked in sequence in a direction away from the fourth surface **112**.

Example 5

[0095] The configuration of example 5 is the same as in example 1, except that the outer glass plate **15** is a grey glass with the thickness of 2.1 mm and has a visible light transmittance of 28%, the adhesive layer **17** is a grey PVB film with the thickness of 0.76 mm and has a visible light transmittance of 18%, and the low-emissivity coating **13** includes a Si_3N_4 layer (6 nm), a SiO_2 layer (63 nm), an ITO layer (20 nm), a SiO_2 layer (34 nm), an ITO layer (100 nm), a Si_3N_4 layer (18 nm), and a SiO_2 layer (82 nm) stacked in sequence in a direction away from the fourth surface **112**.

[0096] Performance testing is as follows.

[0097] For each of the laminated glass **10** of examples 1 to 5 obtained through vehicle glass production process, a visible light transmittance (TL) test, an infrared transmittance (T_{nIR}) test, an infrared reflectivity (R_{IR}) test, and a total solar transmittance (TTS) test are performed, and the thermal insulation index C is calculated. Testing results and calculation results of examples 1 to 5 are shown in table 1.

[0098] Visible light transmittance (TL) test: the transmittance of the laminated glass **10** for visible light with a wavelength of 380 nm to 780 nm is measured and calculated according to ISO9050.

[0099] Infrared transmittance (T_{nIR}) test: the transmittance of the laminated glass **10** for infrared with a wavelength of 1000 nm to 1250 nm is measured and calculated according to ISO9050.

[0100] Infrared reflectivity (R_{IR}) test: the reflectivity of the laminated glass **10** for infrared with a wavelength of 780 nm to 2500 nm is measured at one side of the low-emissivity coating **13** away from the inner glass plate **11** and calculated according to ISO9050.

[0101] Total solar transmittance (TTS) test: the total solar transmittance of the laminated glass **10** for light with a wavelength of 300 nm to 2500 nm is measured and calculated according to ISO9050.

[0102] Calculation of thermal insulation index C: the thermal insulation index of the laminated glass **10** is calculated through formula $C=T_{nIR}/R_{IR}$.

[0103] Emissivity e test: an emissivity meter is used to measure the laminated glass **10** at one side of the low-emissivity coating **13** away from the inner glass plate **11**, that is, at the inside of the vehicle.

TABLE 1

measurement results and calculation results of laminated glass 10 of examples 1 to 5					
	Example 1	Example 2	Example 3	Example 4	Example 5
Mass fraction of total iron as Fe_2O_3 in the inner glass plate	1.81%	1.82%	1.83%	1.84%	1.84%
Total thickness of the ITO layer in the low-emissivity coating	140 nm	140 nm	140 nm	100 nm	120 nm
Ratio of a mass of total iron as Fe_2O_3 in the inner glass plate to a mass of indium in the 1 to 3 transparent conductive oxide coatings	127.4	128.1	128.8	181.4	151.1
Visible light transmittance TL	0.67%	3.17%	1.40%	6.90%	5.08%
Infrared transmittance T_{nIR}	0.28%	0.46%	0.26%	1.54%	0.82%
Infrared reflectivity R_{IR}	5.90%	7.20%	6.75%	6.98%	6.98%
Total solar transmittance TTS	17.40%	17.98%	17.67%	21.34%	19.73%
Thermal insulation index C	0.047	0.064	0.039	0.221	0.117
Emissivity e	0.17	0.17	0.17	0.19	0.18

[0104] According to table 1, by selecting appropriate outer glass plate **15**, adhesive layer **17**, and inner glass plate **11** with the low-emissivity coating **13**, the visible light transmittance of the laminated glass **10** provided in examples 1 to 5 may be within the range of 0.5% to 7%, or even within the range of 0.5% to 4%, the total solar transmittance of the laminated glass **10** provided in examples 1 to 5 may be less than or equal to 25%, or even less than or equal to 20%, and the emissivity of the laminated glass **10** provided in examples 1 to 5 may be less than or equal to 0.20. In this

way, the laminated glass **10** provided in examples 1 to 5 each has a low visible light transmittance, an excellent heat insulation, and a low emissivity, and is more suitable for use as sunroof glass or panoramic roof glass. Meanwhile, the laminated glass **10** provided in examples 1 to 5 each has a relatively low thermal insulation index *C* in a range of 0.03 to 0.25, thereby having a good thermal blocking effect. When the laminated glass **10** provided in examples 1 to 5 are used as sunroof glass or panoramic roof glass, a good body surface sensation may be provided for passengers.

Examples 6 to 10

[0105] In examples 6 to 10, the laminated glass **10** each includes the low-emissivity coating **13**, the inner glass plate **11**, the adhesive layer **17**, the infrared reflection coating **18**, and the outer glass plate **15** stacked in sequence. The low-emissivity coating **13** includes a Si_3N_4 layer, a SiO_2 layer, an ITO layer, a SiO_2 layer, an ITO layer, a Si_3N_4 layer, and a SiO_2 layer stacked in sequence on the fourth surface **112** of the inner glass plate **11** through magnetron sputtering process, where the ITO layer is the transparent conductive oxide coating and the others are the first dielectric layers. The low-emissivity coating **13** includes two ITO layers, and the target material for magnetron sputtering the ITO layer has a mass ratio of In_2O_3 to SnO_2 of 90:10. The infrared reflection coating **18** is deposited on the second surface **152** of the outer glass plate **15** through magnetron sputtering process, and the infrared reflection coating **18** includes two or three metal layers **184** and multiple second dielectric layers, where the metal layer **184** is a silver layer.

Example 6

[0106] The inner glass plate **11** is a grey glass with the thickness of 2.1 mm and has a visible light transmittance of 28%. The adhesive layer **17** is a grey PVB film with the thickness of 0.76 mm and has a visible light transmittance of 18%. The outer glass plate **15** is a clear glass with the thickness of 2.1 mm and has a visible light transmittance of 88%. The low-emissivity coating **13** includes a Si_3N_4 layer (6 nm), a SiO_2 layer (63 nm), an ITO layer (20 nm), a SiO_2 layer (34 nm), an ITO layer (120 nm), a Si_3N_4 layer (18 nm), and a SiO_2 layer (82 nm) stacked in sequence in a direction away from the fourth surface **112**. The infrared reflection coating **18** includes a Si_3N_4 layer (32 nm), an AZO layer (7 nm), an Ag layer (10.1 nm), a TiO_x layer (2 nm), an AZO layer (8 nm), a ZnSnO_x layer (66 nm), an AZO layer (7 nm), an Ag layer (8.1 nm), a TiO_x layer (2 nm), an AZO layer (8 nm), a ZnSnO_x layer (145 nm), and a Si_3N_4 layer (5 nm) stacked in sequence in a direction away from the second surface **152**, where the AZO is aluminum-doped zinc oxide.

Example 7

[0107] The configuration of example 7 is the same as in example 6, except that the adhesive layer **17** is a grey PVB film with the thickness of 0.76 mm and has a visible light transmittance of 8%.

Example 8

[0108] The configuration of example 8 is the same as in example 6, except that the adhesive layer **17** is a grey PVB film with the thickness of 0.76 mm and has a visible light transmittance of 2%.

Example 9

[0109] The configuration of example 9 is the same as in example 6, except that the inner glass plate **11** is a grey glass with the thickness of 2.1 mm and has a visible light transmittance of 40%, the adhesive layer **17** is a grey PVB film with the thickness of 0.76 mm and has a visible light transmittance of 8%, and the infrared reflection coating **18** includes a ZnSnO_x layer (26 nm), an AZO layer (10.8 nm), an Ag layer (13 nm), an AZO layer (8.6 nm), a ZnSnO_x layer (56.8 nm), an AZO layer (8.8 nm), an Ag layer (14.4 nm), an AZO layer (8.4 nm), a ZnSnO_x layer (54.8 nm), an AZO layer (8.6 nm), an Ag layer (13 nm), an AZO layer (9 nm), a ZnSnO_x layer (19.9 nm), and a Si_3N_4 layer (10.3 nm) stacked in sequence in a direction away from the second surface **152**.

Example 10

[0110] The configuration of example 10 is the same as in example 6, except that the adhesive layer **17** is a grey PVB film with the thickness of 0.76 mm and has a visible light transmittance of 5%, and the infrared reflection coating **18** includes a ZnSnO_x layer (26 nm), an AZO layer (10.8 nm), an Ag layer (13 nm), an AZO layer (8.6 nm), a ZnSnO_x layer (56.8 nm), an AZO layer (8.8 nm), an Ag layer (14.4 nm), an AZO layer (8.4 nm), a ZnSnO_x layer (54.8 nm), an AZO layer (8.6 nm), an Ag layer (13 nm), an AZO layer (9 nm), a ZnSnO_x layer (19.9 nm), and a Si_3N_4 layer (10.3 nm) stacked in sequence in a direction away from the second surface **152**.

[0111] Performance testing is as follows.

[0112] For each of the laminated glass **10** of examples 6 to 10 obtained through vehicle glass production process, a visible light transmittance (TL) test, an infrared transmittance (T_{nIR}) test, an infrared reflectivity (R_{IR}) test, and a total solar transmittance (TTS) test are performed, and the thermal insulation index *C* is calculated. Testing results and calculation results of examples 6 to 10 are shown in table 2.

[0113] Visible light transmittance (TL) test: the transmittance of the laminated glass **10** for visible light with a wavelength of 380 nm to 780 nm is measured and calculated according to ISO9050.

[0114] Infrared transmittance (T_{nIR}) test: the transmittance of the laminated glass **10** for infrared with a wavelength of 1000 nm to 1250 nm is measured and calculated according to ISO9050.

[0115] Infrared reflectivity (R_{IR}) test: the reflectivity of the laminated glass **10** for infrared with a wavelength of 780 nm to 2500 nm is measured at one side of the low-emissivity coating **13** away from the inner glass plate **11** and calculated according to ISO9050.

[0116] Total solar transmittance (TTS) test: the total solar transmittance of the laminated glass **10** for light with a wavelength of 300 nm to 2500 nm is measured and calculated according to ISO9050.

[0117] Calculation of thermal insulation index *C*: the thermal insulation index of the laminated glass **10** is calculated through formula $C=T_{nIR}/R_{IR}$.

[0118] Emissivity *e* test: an emissivity meter is used to measure the laminated glass **10** at one side of the low-emissivity coating **13** away from the inner glass plate **11**, that is, at the inside of the vehicle.

TABLE 2

measurement results and calculation results of laminated glass 10 of examples 6 to 10					
	Example 6	Example 7	Example 8	Example 9	Example 10
Mass fraction of total iron as Fe_2O_3 in the inner glass plate	1.81%	1.82%	1.83%	1.18%	1.84%
Total thickness of the ITO layer in the low-emissivity coating	140 nm	140 nm	140 nm	140 nm	140 nm
Ratio of a mass of total iron as Fe_2O_3 in the inner glass plate to a mass of indium in the 1 to 3 transparent conductive oxide coatings	127.4	128.1	128.8	83.08	129.5
Visible light transmittance TL	5.22%	2.53%	0.52%	2.74%	1.21%
Infrared transmittance T_{nIR}	0.68%	0.13%	0.09%	0.07%	0.06%
Infrared reflectivity R_{IR}	6.54%	7.57%	5.90%	6.20%	6.10%
Total solar transmittance TTS	15.45%	13.04%	12.65%	11.66%	10.98%
Thermal insulation index C	0.104	0.017	0.015	0.011	0.011
Emissivity e	0.17	0.17	0.17	0.17	0.17

[0119] According to table 2, by adding the infrared reflection coating 18 and selecting appropriate adhesive layer 17 and inner glass plate 11 with the low-emissivity coating 13, the visible light transmittance of the laminated glass 10 provided in examples 6 to 10 may be within the range of 0.5% to 6%, or even within the range of 0.5% to 3%, the total solar transmittance of the laminated glass 10 provided in examples 6 to 10 may be less than or equal to 16%, or even less than or equal to 13%, and the emissivity of the laminated glass 10 provided in examples 6 to 10 may be less than or equal to 0.20. In this way, the laminated glass 10 provided in examples 6 to 10 each has a low visible light transmittance, an excellent heat insulation, and a low emissivity, and is more suitable for use as sunroof glass or panoramic roof glass. Meanwhile, the laminated glass 10 provided in examples 6 to 10 each has a relatively low thermal insulation index C in a range of 0.009 to 0.15, or even 0.009 to 0.10, or more even 0.009 to 0.05, thereby having a good thermal blocking effect. When the laminated glass 10 provided in examples 6 to 10 are used as sunroof glass or panoramic roof glass, a good body surface sensation may be provided for passengers.

[0120] Reference is made to FIG. 11. Embodiments of the disclosure further provide a vehicle 30. The vehicle 30 includes a vehicle body 31 and the laminated glass 10 provided in the disclosure. The laminated glass 10 is mounted on the vehicle body 31 as at least one of a sunroof glass, a side window glass, or a rear windshield glass of the vehicle 30. In the vehicle 30 provided in the disclosure, the visible light transmittance and the total solar transmittance of the laminated glass 10 are both low, so that the laminated glass 10 has a good thermal blocking effect. In addition, the

laminated glass 10 has a low emissivity, so that passengers can have a good experience when the laminated glass 10 is applied to the vehicle.

[0121] Optionally, the vehicle 30 may be, but is not limited to a car, a truck, a sedan, etc.

[0122] It may be understood that, in some embodiments, the sunroof glass of the vehicle 30 is the laminated glass 10 provided in the disclosure. In other embodiments, the side window glass of the vehicle 30 is the laminated glass 10 provided in the disclosure. In other embodiments, the rear windshield glass of the vehicle 30 is the laminated glass 10 provided in the disclosure. In other embodiments, the sunroof glass and the rear windshield glass of the vehicle 30 are laminated glass 10 provided in the disclosure. In other embodiments, the sunroof glass and the side window glass of the vehicle 30 are laminated glass 10 provided in the disclosure. In other embodiments, the side window glass and the rear windshield glass of the vehicle 30 are laminated glass 10 provided in the disclosure. In other embodiments, the sunroof glass, the side window glass, and the rear windshield glass of vehicle 30 are the laminated glass 10 provided in the disclosure.

[0123] The term “embodiment” or “implementation” referred to herein means that a particular feature, structure, or feature described in conjunction with the embodiment may be contained in at least one embodiment of the disclosure. Phrases of “embodiment” appearing in various places in the specification does not necessarily refer to the same embodiment, nor does it refer to an independent or alternative embodiment that is mutually exclusive with other embodiments. It is explicitly and implicitly understood by those of ordinary skill in the art that an embodiment described herein may be combined with other embodiments. In addition, it can be further understood that the features, structures, or characteristics described in embodiments of the disclosure may be combined arbitrarily without contradiction therebetween to form another embodiment that does not depart from the spirit and scope of the technical solutions of the disclosure.

[0124] Finally, it should be noted that the foregoing embodiments are merely intended to illustrate but not limit the technical solutions of the disclosure. Although the disclosure is described in detail with reference to the foregoing optimal embodiments, those of ordinary skill in the art should understand that modifications or equivalent replacements can be made to the technical solutions of the disclosure without departing from the spirit and scope of the technical solutions of the disclosure.

1. A laminated glass, comprising an outer glass plate, an adhesive layer, an inner glass plate, and a low-emissivity coating stacked in sequence; wherein

the outer glass plate has a first surface and a second surface opposite to each other, the inner glass plate has a third surface and a fourth surface opposite to each other, and the adhesive layer is used to connect the second surface and the third surface;

the low-emissivity coating is disposed on the fourth surface of the inner glass plate and comprises 1 to 3 transparent conductive oxide coatings; and

a thermal insulation index C of the laminated glass is calculated according to the equation: $C = T_{nIR} / R_{IR}$, wherein a range of the thermal insulation index C is $0.008 \leq C \leq 0.4$, T_{nIR} is a transmittance of the laminated glass for infrared with a wavelength of 1000 nm to

1250 nm, and R_{IR} is a reflectivity of the laminated glass for infrared with a wavelength of 780 nm to 2500 nm measured at one side of the low-emissivity coating away from the inner glass plate.

2. The laminated glass of claim 1, wherein a material of each of the 1 to 3 transparent conductive oxide coatings is indium tin oxide (ITO), the inner glass plate comprises iron oxide, and a ratio of a mass of total iron as Fe_2O_3 in the inner glass plate to a mass of indium in the 1 to 3 transparent conductive oxide coatings ranges from 20:1 to 500:1.

3. The laminated glass of claim 2, wherein the inner glass plate is a tinted glass, a mass fraction of total iron as Fe_2O_3 in the inner glass plate ranges from 0.9% to 2.2%, and a thickness of the inner glass plate ranges from 0.7 mm to 2.1 mm.

4. The laminated glass of claim 2, wherein a total thickness of the 1 to 3 transparent conductive oxide coatings ranges from 50 nm to 300 nm, and a ratio of a mass of total iron as Fe_2O_3 in the inner glass plate to a mass of indium in the 1 to 3 transparent conductive oxide coatings ranges from 50:1 to 200:1.

5. The laminated glass of claim 1, wherein a visible light transmittance of the laminated glass ranges from 0.5% to 10%, an emissivity of the laminated glass measured at the side of the low-emissivity coating away from the inner glass plate is less than or equal to 0.25, and a visible light reflectivity of the laminated glass measured at the side of the low-emissivity coating away from the inner glass plate is less than or equal to 4%.

6. The laminated glass of claim 1, wherein the low-emissivity coating further comprises at least two first dielectric layers, a first dielectric layer is disposed on each of two opposite surfaces of each transparent conductive oxide coating, and a material of the at least two first dielectric layers is selected from nitride, oxide, or oxynitride of at least one of zinc (Zn), tin (Sn), titanium (Ti), silicon (Si), aluminum (Al), magnesium (Mg), or zirconium (Zr).

7. The laminated glass of claim 1, further comprising:
an infrared reflection coating, wherein the infrared reflection coating is disposed between the inner glass plate and the outer glass plate, the infrared reflection coating comprises 1 to 5 metal layers, and a material of each of the 1 to 5 metal layers is selected from a metal or an alloy of at least one of silver (Ag), aurum (Au), copper (Cu), aluminum (Al), or platinum (Pt).

8. The laminated glass of claim 7, wherein the infrared reflection coating further comprises at least two second dielectric layers, a second dielectric layer is disposed on each of two opposite surfaces of each metal layer, and a material of the at least two second dielectric layers is selected from nitride, oxide, or oxynitride of at least one of zinc (Zn), tin (Sn), titanium (Ti), silicon (Si), Al, nickel (Ni), chromium (Cr), niobium (Nb), magnesium (Mg), zirconium (Zr), gallium (Ga), yttrium (Y), indium (In), antimony (Sb), vanadium (V), or tantalum (Ta).

9. The laminated glass of claim 7, wherein the outer glass plate is a clear glass with a visible light transmittance greater than or equal to 80%.

10. The laminated glass of claim 1, wherein the laminated glass does not comprise an infrared reflection coating, the outer glass plate is a tinted glass, a mass fraction of total iron as Fe_2O_3 in the outer glass plate ranges from 0.7% to 2.2%, and a thickness of the outer glass plate ranges from 2.1 mm to 4.2 mm.

11. The laminated glass of claim 1, wherein the range of the thermal insulation index C of the laminated glass is $0.03 \leq C \leq 0.25$.

12. The laminated glass of claim 1, wherein the range of the thermal insulation index C of the laminated glass is $0.009 \leq C \leq 0.15$.

13. The laminated glass of claim 1, wherein the adhesive layer is a tinted thermoplastic polymer film, and the thermoplastic polymer film is selected from at least one of polyvinyl butyral, polyurethane, ethylene-vinyl acetate copolymer, or ionic polymer, and the adhesive layer has a visible light transmittance ranging from 1% to 20%.

14. The laminated glass of claim 1, wherein a visible light transmittance of the inner glass plate ranges from 34% to 42%, and a value range of L in a Lab value of a color of the inner glass plate is 65 to 71.5, a value range of a is -3.5 to -2 , and a value range of b is 2 to 3.5.

15. The laminated glass of claim 1, wherein a visible light transmittance of the inner glass plate ranges from 26% to 34%, and a value range of L in a Lab value of a color of the inner glass plate is 58.5 to 65, a value range of a is -5 to -3.5 , and a value range of b is 0.5 to 2.

16. The laminated glass of claim 1, further comprising:
a light-adjustment component disposed between the second surface and the third surface, and the light-adjustment component comprises at least one of a polymer-dispersed liquid crystal light-adjustment film, a suspended particle light-adjustment film, or an electrochromic light-adjustment film.

17. The laminated glass of claim 1, wherein the inner glass plate comprises 1.1% to 1.9% of total iron as Fe_2O_3 , 140 ppm to 300 ppm of Co_2O_3 , 10 ppm to 320 ppm of Cr_2O_3 , 55 ppm to 75 ppm of SrO, and 10 ppm to 30 ppm of Se.

18. The laminated glass of claim 1, wherein a visible light transmittance of the laminated glass ranges from 0.5% to 6%, an emissivity of the laminated glass measured at the side of the low-emissivity coating away from the inner glass plate is less than or equal to 0.20, and a total solar transmittance of the laminated glass is less than or equal to 16%.

19. The laminated glass of claim 1, wherein the range of the thermal insulation index C of the laminated glass is $0.009 \leq C \leq 0.10$; or the range of the thermal insulation index C of the laminated glass is $0.009 \leq C \leq 0.05$.

20. A vehicle, comprising:

a vehicle body; and

a laminated glass comprising an outer glass plate, an adhesive layer, an inner glass plate, and a low-emissivity coating stacked in sequence, wherein the laminated glass is mounted on the vehicle body as at least one of a sunroof glass, a side window glass, or a rear windshield glass of the vehicle; wherein

the outer glass plate has a first surface and a second surface opposite to each other, the inner glass plate has a third surface and a fourth surface opposite to each other, and the adhesive layer is used to connect the second surface and the third surface;

the low-emissivity coating is disposed on the fourth surface of the inner glass plate and comprises 1 to 3 transparent conductive oxide coatings; and

a thermal insulation index C of the laminated glass is calculated according to the equation: $C = T_{nIR} / R_{IR}$, wherein a range of the thermal insulation index C is $0.008 \leq C \leq 0.4$, T_{nIR} is a transmittance of the laminated

glass for infrared with a wavelength of 1000 nm to 1250 nm, and R_{IR} is a reflectivity of the laminated glass for infrared with a wavelength of 780 nm to 2500 nm measured at one side of the low-emissivity coating away from the inner glass plate.

* * * * *